

Clean Drinking Water Distribution Network

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An important utility in the world is water. Water is used in a variety of different processes and for many reasons. To manage the water delivery in an area, water utility networks are implemented. These can be for water delivery, storm water management, fire hydrants and grey or black water. With an increasing demand for water and growing populations, it is important for clean water distribution networks to buildings and households to be created.

The conceptual utility network I have created focuses only on the distribution of clean drinking water, from the source to the household, for a small neighbourhood. The general flow of the network will be from Source → Treatment Plant → Holding Tank (Reservoir) → Water Mains → Service Lines → Drinking Water in the buildings. This network will also contain, pressurizers (pumps), joints, valves and smaller distribution lines to safely deliver the water to the buildings without contamination. For this network, since it will be treated water, there will be a strong importance on quality control, checks and monitoring to ensure there are no unknown breaks or areas where contaminants can enter the water supply.

Data Collection

Before creating the water distribution network, data will need to be collected to properly understand the area which the network will service and all elements that will impact its design. Data on the geography and geology of the area, the current infrastructure, population of the area and data on the potential users will be required.

Topography will be important to analyze since it will affect how the water will be pumped through the water mains and where it will be located. Elevation data will be used to determine how many pumps and how much pressure will be needed to move water through the mains. Uphill will require more pressure versus downhill. This data can be obtained through physical maps or digital maps/analysis using aerial photographs of the area, combining both past year's data with survey's done specifically for this project. Contour lines will be important to look along with slope data and a digital elevation model (DEM).

Information collected regarding the location of lakes, rivers, runoff area, marshes, bogs, ground water and aquifers will also be important since they can be evaluated to become a potential water source or could impact the water quality of the area in the long run. This data can also be collected through surveys, past physical and digital maps, orthophotos as well as through remote sensing data. Analysis can be done on remotely sensed or GIS data to isolate the water in the study area, classify it, and eliminate the features that are not potential sources or in acceptable areas.

Geological data will also need to be collected from a combination of past and present physical and/or digital maps. The geological data will provide information on earth's layers beneath the surface and its makeup. This will provide information on if there are any soils or rocks that will impact placing the water main materials under ground, for example will the rock be too hard to drill through, could the soils erode quickly if they come in contact with high volumes of water as well as information on potential issues with water quality, for example, are there components of the soils and rocks that will affect the water quality, are there pre existing aquifers or ground water streams that could flow into the water source or reservoir contaminating the water further. This data can be compiled through past data, soil samples, water flow tests (ex. tracking water flow using dyes) and geochemical analysis.

Data on previous environmental disasters for the area will need to be compiled. Landslides, earthquakes, flooding which could impact the network infrastructure, especially with the potential for displacing the mains, treatment centre and impacting water quality can be compiled through past records, studies, and prepared maps, looking at causes, frequency, damage, as well as recommendations for how to minimize.

Land use data will also want to be gathered to roughly determine the areas that will require more water, for example looking at commercial areas versus neighbourhoods, and what may impact the construction of the network, for example protected land, major roadways, current underground infrastructure. This data can be collected through past surveys of the area, looking at data from past construction projects, orthophotos as well as land use mapping (physical or digital). This data would also have to be confirmed with the city or governing body's records the area falls under. Data on other underground networks, such as storm water or sewage, should be obtained so that when constructing the water network, those are not disturbed, and the water source chosen is not close to their output locations.

Population data will need to be analyzed to determine who needs the water and how much water will be needed. Past surveys and maps containing information on water consumption levels as well as ones for population density should be studied. More current population data can be collected by surveys and used along with past data can be used to create projected future population values for the area which can impact how much water the network will need to be able to handle.

Proposed Drinking Water Distribution Network

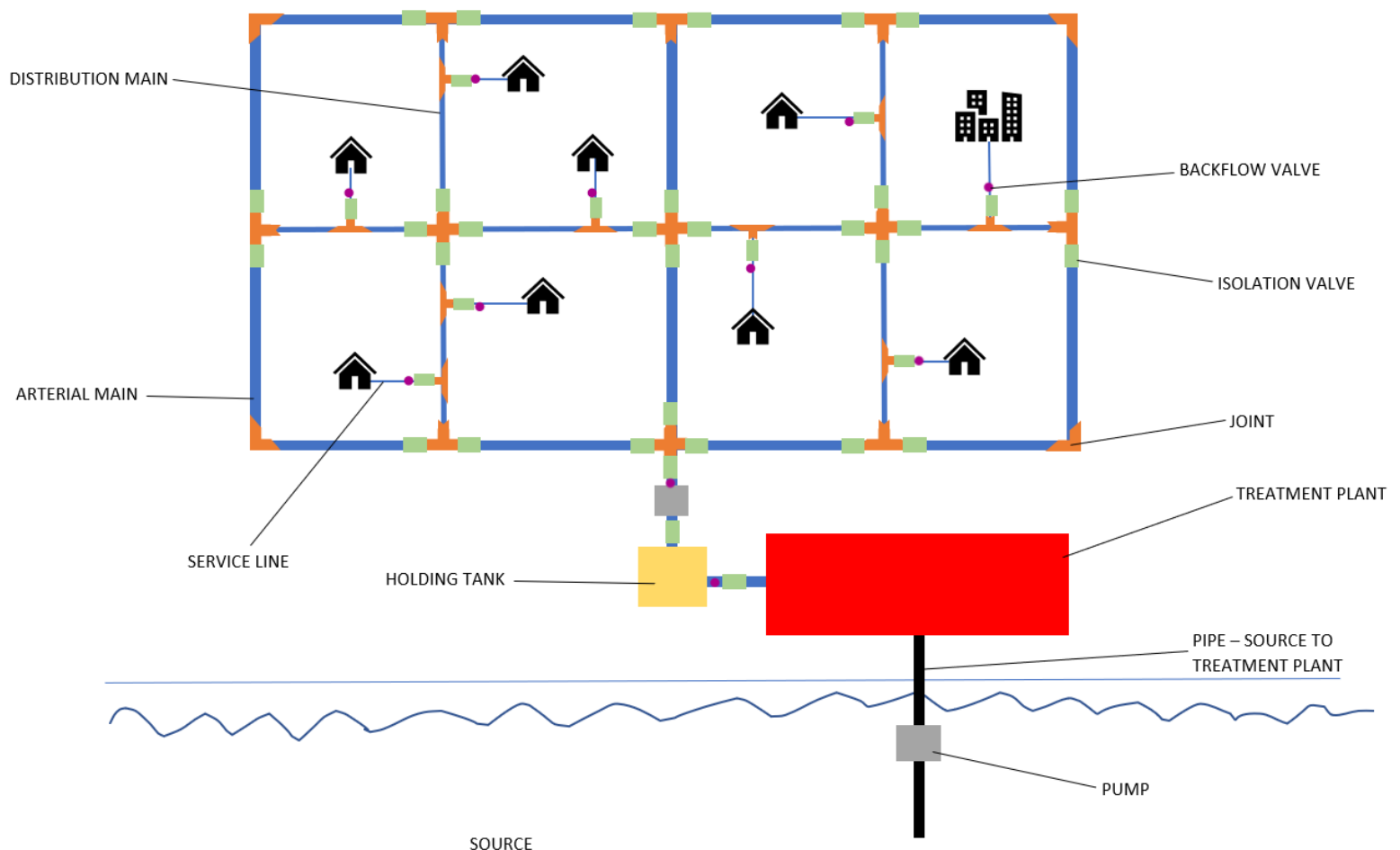


Figure 1: Proposed clean drinking water distribution network.

Features and Attributes

Source

The source will need to contain enough water to be able to supply the buildings along the network [1]. To be eligible to become the source it will need to not contain any harmful chemicals or elements that cannot be treated. For this network the source will be a lake. The water will be drawn from the lake through pipes that will run into the centre at depth. This water at depth contains less micro-organisms, organic matter, and debris than surface water due to less interaction with humans and animals, which will minimize the amount of treatment the water will need to go through [1].

Attributes associated with source can include:

- Source Type
 - Will let the user know what type of source it is. For example, is it a lake, well, or aquifer.
- General source Location
 - Will let user know generally where it is located.
- Coordinates
 - This will provide a more detailed and precise location of the source.
- Volume
 - This value will show how much water the source contains. This value can be used when determining how much water is being used and if this source is still ok to use, for example, it is drying up or replenishing.
- Temperature
 - Will let the user know the general temperature of the water in the source, this will be helpful when deciding what treatment and purification steps the water will need to go through.
- Water quality
 - This will let the user know what the contaminants and element levels within the water are when it is taken from the source before treatment.
- Water clarity
 - This value will provide a number on a predetermined clarity scale for the water sampled from the source. This can be used to decide which treatment processes the water will go through.
- Location of the pipe in the lake
 - This will inform the user where the pipe that removes the water from the source is located.
- Depth of pipe in lake
 - This will identify the depth the pipe removing the water is at in the source.

Treatment Plant

The treatment plant will take the lake water and purify it to acceptable levels for consumption. This can include, water softening, filtration of rock debris, organic matter, dissolved material, desalination and disinfection (to kill and parasites and bacteria) [1, 3]. The process will begin with collection, coagulation, flocculation before moving on to

sedimentation, filtration and finally disinfection [2]. The coagulation step will see the addition of positively charged chemicals to the water. This will neutralize any dirt or dissolved particles that are in the water, allowing them to bind together, creating larger particles. Flocculation then mixes the water to allow for the formation of larger, heavier particles [2]. Next, the water will go through sedimentation. This will separate out the larger, heavier particles and solids, such as soils [2] by having the water become still, with gravity pulling the particles down to the bottom. Filtration will then take the clear water from the surface of the sedimentation process and pass it through multiple filters to remove any remaining particles [2]. Finally, the water will be disinfected by the addition of chemicals, such as chlorine, to kill any bacteria and germs [2].

Attributes associated with treatment plant can include:

- Sample Number
 - This number will identify which sample you are studying.
- Location/Facility ID
 - This attribute will let the user know which treatment facility the sample came from (if there are more than one in your network).
- Filter Size 1
 - If this is filled out yes or no, it will identify which filter the water was filtered through.
- Filter Size 2
 - If this is filled out yes or no, it will identify which filter the water was filtered through.
- Filter Size 3
 - If this is filled out yes or no, it will identify which filter the water was filtered through.
- Disinfectant Chemical Added
 - This will identify the chemicals that were added to the water. This is an important attribute as it will let the user know if the sample went through the disinfection stage.
- Amount of Disinfectant Chemical Added
 - This will let the user know how much of the element or chemical was added and can be used to know the levels within the water and used to determine if too much or too little was added.
- Final water quality
 - This will show if the quality of the water at the end of the treatment is acceptable, identifying which chemicals were added to the water. The results can be compared to a control test/sample and then the level can be indicated here.
- Final water clarity
 - This value will provide a number on a predetermined clarity scale for the water sampled from the treatment plant. This can be used to decide if the water is acceptable to enter the network or will need to be treated again.
- Last cleaned

- This date will let the user know when the treatment plant was last cleaned.

Holding Tanks/Reservoir

The holding tank, or reservoir, will be the location where the treated water is stored before it enters the pipe network. Holding tanks or reservoirs can be open or closed, with closed being the preferred option as it minimized external contaminants or debris from entering the waters. These can be made up of concrete, metal, or PVC plastics. In this network, it will be a closed reservoir, to minimize external particles entering the treated waters, as well as elevated to allow for gravity to push the water into the network.

Attributes associated with holding tank/reservoir can include:

- Holding Tank/Reservoir ID
 - This number will be used to identify which holding tank the user is working with.
- Coordinates
 - This attribute will provide the exact location of the holding tank.
- Capacity
 - This will let the user know how much water the holding tank can hold.
- Water Volume
 - This value will identify how much water is currently in the tank. This value along with the capacity, can be used to determine if there is still space or if it is full.
- Weight – No water
 - This will let the user know what the weight of the holding tank is without water.
- Weight – With Water
 - This will let the user know what the weight of the holding tank is with water.
- Materials
 - This will let the user know the main material that the holding tank is made out of.
- Installation Date
 - This inform the user of when the holding tank was made and how old it is. This value compared to the guidelines for holding tank life spans can be used to determine when and if it will need replacing.
- Water quality
 - This Identifies the elements in the water in the holding tank. The results can be compared to a control test/sample and then the level can be indicated here.
- Water clarity
 - This value will provide a number that can be used on a predetermined clarity scale for the water sampled from the holding tank.
- Last Cleaned
 - This date will let the user know when it was last cleaned and when it will need to be cleaned next. If there are issues with water quality or clarity,

the date it was last cleaned could be a factor as to why it is not up to standard (ex. algae growing could mean it has not been cleaned in a long time).

Pumps

Multiple pumps can be found within the network to ensure that the water pressure and water movement remains consistent through out. A low-lift pump, one that works at low pressures to move large amounts of water, will be used to get the water from the source to the treatment plant while a high-lift pump will move the water from the reservoir to the pipe system (arterial mains), operating at higher pressures [3]. Water pumps can also be found along the network to maintain water flow and pressure. These pumps can use rotating impellers to move the water, adding energy to the water and raising the pressure [3].

Attributes associated with pumps can include:

- Pump ID
 - This number will identify which pump the user is looking at.
- General Location
 - The general location, indicating an intersection or street name and degree can be used to locate the pump.
- Coordinates
 - This will provide the exact location of the pump.
- Pump Type
 - This will indicate to the user what type of pump is at this location, for example, if it is a low lift or high lift pump.
- Pressure
 - This will let the user know how strong the pump is .
- Rotation Speed
 - This will identify the speed at which the impeller or rotor is moving at.
- Installation Date
 - This will let the user know when the pump was installed.
- Maintenance Date
 - This will provide information on when the pump was last inspected.
- Last Cleaned
 - This date will let the user know when the pump was last cleaned as a dirty pump could affect the water quality.

Water Mains

Water mains will be used to transport the water through the network. These will be pipes of varying sizes and lengths depending on their location. They will be underground and connected in a loop pattern with smaller loops branching off to distribute water evenly to the buildings. As a loop, the water will be continuously flowing, eliminating dead ends which can cause buildups of water and slower flow rates [1]. The arterial mains will carry the majority of the water through the network. The distribution mains branch off from the arterial main, carrying less water, in a grid like system. This will allow for water to still be carried to other areas of the network if one section has a break, a leak or is isolated for

testing. Service lines branch off from the distribution mains, carrying the water to the buildings. These pipes can be made from a variety of materials, cast iron or ductile and polyvinyl chloride (PVC) are some of the top choices [4]. The materials are chosen to minimize the amount of corrosion from the pipe experiences from the water, reduce the growth of organisms and they are able to withstand high pressures, both from the water and the weight of the earth on top of them [1].

Attributes associated with the water mains can include:

- Identification Number
 - This number will be the technical identifier for the pipe the user is working with.
- General Location
 - This will be the general location of the pipe. It can be listed as a street corner or which section of the network it can be found in.
- Coordinates
 - This will be the exact location of the water main.
- Type of water main
 - This will indicate which type of main it is apart of, the arterial main, distribution main or service line.
- Material
 - This will let the user know what it is made out of.
- Length
 - This will let the user know how long the main will be.
- Width
 - This will let the user know how wide the main will be.
- Height
 - This will let the user know how high the main will be.
- Depth Underground
 - This will indicate how far beneath the surface the water main is and the distance to the top of the pipe.
- Soil Type above pipe
 - This will let the user know what is above the water main. For example, if it is organism rich soil, sand, silt or rock.
- Amount of water moving through the pipe
 - This will let the user know the average quantity of water that flow through this water main.
- Installation Date
 - The installation date can be used by the user to determine how old the water main is and if it will need to be replaced soon.
- Last Checked/Tested
 - This date will let the user know when the last maintenance and cleaning was done on this section of the water main.

Joints

Joints will be used to connect the pipes to each other, allowing for a continuous network. Joints can be made out of similar materials as the mains and in a variety of shapes and sizes to ensure the network can move in any direction where necessary.

Attributes associated with the joints can be:

- Identification Number
 - This will be the technical identifier of the joint in the network.
- Type of Joint
 - This will identify for the user what type of joint it is. For example, if it is a corner joint or a three-way connector.
- Coordinates
 - This will identify the exact location of the joint.
- Installation Date
 - This date will let the user know when it was first installed.
- Depth
 - This will inform the user how far beneath the surface the joint can be found.
- Connection on Left
 - This will identify the piece that the joint is connected to on the left, for example, it will let the user know if it connected to a valve or a main (indicating the identification number and type).
- Connection on Right
 - This will identify the piece that the joint is connected to on the right, for example, it will let the user know if it connected to a valve or a main (indicating the identification number and type).
- Last Checked/Tested
 - This date will let the user know when it was last checked to determine if it was still up to the standard and installed correctly.

Valves

Within the network there will be multiple valves connected to the water mains. These are used to minimize back flow in the water mains, especially along the service lines from the buildings, as well as used to isolate areas of the water main from the rest of the network. This can be used to stop the flow of water in one area where there are quality issues or a break while water can still flow to the rest of the buildings in the network [1].

Attributes associated with the valves can include:

- Identification Number
 - This will be the technical identifier for the valve in the network.
- Type of Valve
 - This will let the user know which type of valve it is, for example, an isolation valve or back flow valve.
- Which area of the network does it apply to

- This will provide the general location the valve is in apart of the network, identify which areas it will affect.
- Coordinates
 - This will be the exact location of the valve.
- Material
 - This identifies what the valve is made out of.
- Installation date
 - This will let the user know when the valve was installed and when it will need to be replaced next.
- Last checked/tested
 - This will let the user know when the valve was last checked to be in working condition and clean.
- Which way to open the valve
 - This will let an user who is opening or using the valve which way to turn or move the lever to make it work.
- Depth
 - This identifies how far below the surface the top of the valve is .
- Connection on Left
 - This will identify what the valve is connected to on the left. For example, if it is connected to a main or a joint (along with the identification number).
- Connection on Right
 - This identifies what the valve is connected to on the right. For example, if it is connected to a main or a joint (along with the identification number).

Connectivity

Within the network, tight connections between the various pipes and valves will be important to minimize the risk of any contaminants and particles entering the clean water. The features are fitted together snugly by pushing the ends together, with one smaller than the other. Thrust blocks can also be used to prevent the separation of two pieces fit together [3]. Once fit together the flow should be smooth throughout the entire network with strategically placed isolation valves, to be able to stop the flow in certain areas, and backflow valves, to prevent water from moving the wrong direction.

Figure 2 displays the connectivity between the network features from the treatment plant to the arterial mains. The flow of water will be sent through the treatment plant and pumped out through an isolation and backflow valve into the holding tank. The isolation valve and backflow valve are located at this spot to prevent any contaminated water from entering back into the treatment plant if there are issues with the holding tank. It will allow for the treatment plant to be cut off from the rest of the network while maintenance is underway. Alternately, if there is an issue at the treatment plant, it can be isolated from the holding tank. The water is then moved from the holding tank through a pump (if gravity is not used to move the water), then through backflow and isolation valves and finally through the joints connecting this section to the arterial mains (and rest of the network). The back flow valve is used to prevent any water from flowing from the network to the holding tank

while the isolation valve is here to allow for the network to be cut off from the holding tank if there are issue with water quality or ongoing maintenance.

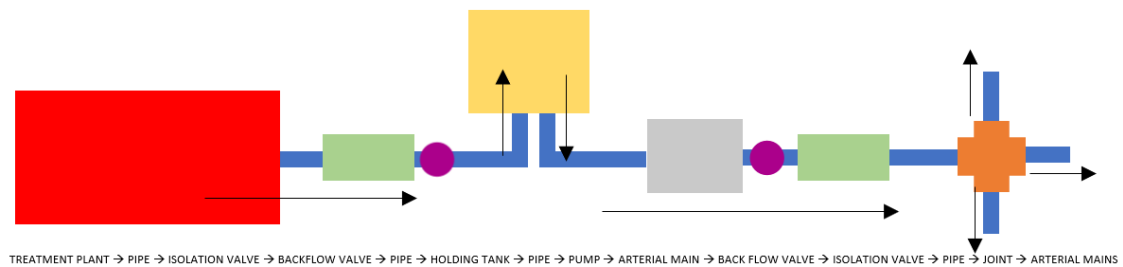


Figure 2: Close up of the connections between the features between the treatment plant and the arterial mains.

From the network to the building, the process is similar. Figure 3 displays the connection of features through the various mains to the house. The water will flow through the arterial main, branching off into a smaller delivery main, joined together by a joint. The water will then flow from the delivery main through another joint, an isolation valve, back flow valve and service line before entering the building. The joints vary in size to connect the different water mains together, preventing any leaks. The isolation valve is placed between the distribution main and service line to be able to cut off the building from the rest of the network. This will be useful in cases where there is construction happening on the building, if there are water quality issues, watermain breaks or leaks. The back flow valve prevents any contaminated water entering the system from the building, such as black water.

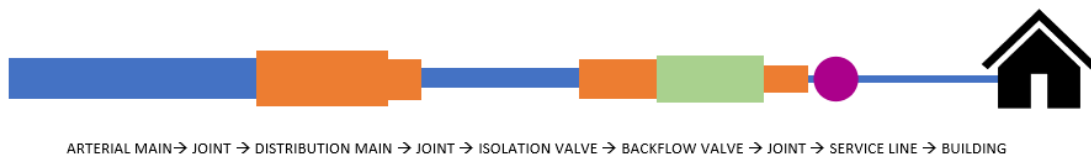


Figure 3: Close up of the connections between the arterial main and the building.

The connectivity is important since at each step of the way there should be minimal opportunities to enter the network and mix with the clean water.

Quality Control

To ensure the clean drinking water network remains in top working condition multiple checks and analyses will be implemented, on both the materials and water quality. The checks should be done periodically, with some dependant on the installation dates, material life span and user feedback.

Water quality is important to monitor closely as it can have severe consequences if contaminated. To successfully monitor, the treatment plant, reservoir and multiple locations along the water mains should be tested at regular intervals, ensuring the entire network is tested, not just one area. Samples of water can be taken and chemically tested for elements that can enter the water from the

corrosion of the metals in the pipes, joints, or valves [1] as well as tested both chemically and visually for the presence of microorganisms, caused by the growth of slime or tubercles [1] within the networks piping. Clarity tests can be done to identify any visual contaminants, such as silt or soils, that have entered the water supply as well as any changes in colour as this could also lead to identifying breaks or cracks in the mains [1]. Both of these tests should be compared to a control sample. A control sample is important so that the proper ratios of chemicals used to treat the water are known and when testing in the future is done, an imbalance in these can be identified, for example chlorine will be added but will need to be monitored so that the concentration levels do not become too high or too low [1].

The flow of water should also be monitored to ensure that it is still travelling in the correct directions and that there is no backflow. This can be done through the shutoff of certain areas and using dye to check the flow direction. Checking the velocity of the water along the network will also be important since, lower flow rates can mean that there is more time that the water is in contact with the pipes which could lead to lower water quality and an increase in metals or organisms within the water supply [1].

Water quality can also indicate locations along the network that may need the pipes, valves or joints replaced. Monitoring for these locations will also need to be done based off of the installation date or modification dates of these pieces. A record of these should be provided so that the lifespan can be monitored and used to know when a section will need to be replaced. A record of the areas that have experienced more trouble with breaks, cracks, leaks, or main movement due to natural disasters should be created and used to identify locations that need to be monitored more closely. Water usage and cost data can be compiled and used to determine areas where there may be a leak. Physical checks along the network can be done to identify areas with raised ground, bulges, cracks in the concrete, warning signs for potential sinkholes, smells, or hissing noises from pipes, which can all be indicators of leaks in the water mains [5]. Thermal imagery can also be used to locate leaks or breaks by looking at the ground surface temperatures. The temperatures will be lower in areas with a leak or break due to the excess water [6]. Another way would be to analyze the NDVI values for the area, but it is less accurate [6].

To be able to check certain sections of the network, the isolation valves can be used to cut off water from certain areas from the rest. These quality control tests should be done during off peak hours to minimize disruption to the users.

Data Presentation, Analysis and Usage

There are multiple ways to visualize the clean water distribution network, for the general public, users, workers and the company who created it. It can be done both physically and digitally, with a mixture of the two being the best option. There are numerous uses for the water network, ranging from academic to practical applications.

Visualisation and Presentation

When collecting the data, most will be a mix of paper and digital. Physical, paper maps, will be helpful to have in case there is an issue with the digital maps created. Some people will also prefer a hard copy. For this water delivery network, there should be a paper map of the entire network, labelled with all the information needed as well as one with more detail

surrounding the treatment plant. General information on the parts, installation dates, control samples and any other relevant information should be printed off and kept in a binder with the date it is current up until boldly stated. This will be used in case of the main, more modern visualization and presentation methods, of a digital map, app, and dashboard, unexpectedly crash. This will also help in the case there is information or records that may be relevant in the future that were not digitized fully.

Web maps, apps and dashboards will be the best and main way to visualize the data. Web maps that display the location of each of the mains (arterial, delivery and service mains), the location of the joints and all valves, treatment plant, holding tanks and source should be created so that all information on the network can be found in one place and manipulated to the view relevant to the users. This web map, combined with a field map, will be used in the monitoring process to help workers in the field understand what they are working with and where they should go. The digital process of using the field app will minimize the risk of lost data due to lost papers or notebooks as well as streamline the data collection. The web map and field application can record all the technical data, including the results on the latest water quality tests and the locations they occurred, locations where leaks have been detected, areas that are being shutoff for maintenance, areas where the pipes are nearing the end of their lifespan, which personnel are working where and the locations where quality control measure will need to take place.

A dashboard would be the best to visualize the data for the users. It is easy to use and navigate and will present all the information regarding the network, however, will not display some of the more technical information which is more important and relevant to the utility company and its workers, than the user as it could cause confusion and take away from the overall goal of the dashboard. An example would be that it may not contain all the information on the water quality tests, like all elements it was tested for, only displaying overall result if the water is up to the standard or not. On the dashboard, on the main map screen, it should contain the network layout, with the attribute information on each piece, such as installation date, whether it is in service or under construction, generalized water quality results at the test locations for that period and the piece or location ID that will pop up when someone clicks on that area. On the side bars of the dashboard, there should be information on the network, its uses, the flow rate and water volume, contact information for people with more questions (and who to reach out to if there is an issue, such as a leak or decrease in water quality), along with information on which areas are currently being worked on or have been shut off. This information should not change when the map scale does. There should also be the data on the number of mains, valves and joints in the area and a graphic on which are working, and which have been shut off. These can be displayed as a bar graph and gauge. Both should change when the map scale changes. The list of features in the map frame should also be displayed, being linked to the bar graph. All data will be updated periodically, with some sections updated daily.

Uses and Users

The data from the water distribution network, displayed in various formats, can be used in many ways by different users. Three users of the data would be the public, an independent construction worker/contractor and an inspector.

The public will want to use this data if their house/building is along this water delivery network. They will want to know when there will be water shutoffs or construction on these pipelines as it can cause major delays and issues. The dashboard and web map can let the user know which sections will be shut off and when (along with the areas currently being worked on). Looking at this data can provide them with an accurate timeframe for when they will need to make other arrangements for water. It also offers a place to report any complications, issues, or strange occurrences with their water. For example, if their water bill has gone up, then they can look to see if any work is being done as that could be a cause, if there is not then they know that there is a possible leak or water main break that they need to report.

A construction worker or contractor that is coming into the neighbourhood to work on building a new house will want to use this data. They will need to gather data on the layout to understand the areas where the pipes are running to make sure that when they break ground, they do not hit a water main. This can be found in the dashboard and web map. The web map may be of more use since it contains more of the technical information relevant to construction. The attribute data on the features, such as age, depth they are at, and the connection types will all be important to know when they are hooking up the new building to the water supply.

An inspector can use this data when they come in to audit the distribution network. They can look at the dashboard and web map to determine if there are any areas that are not up to standard with reporting information to the public and if there is any misinformation. AN auditor can use the recorded water quality test samples to determine if the water treatment process is complaint with the municipal standards or if there is an excess in chemicals. They can use the attribute data, including material and installation date/age, to identify any areas that need to be upgraded as well as confirm that the water quality and composition remains the same through out the network.

Data Analysis

Collecting the data from this water distribution network can be applied to analyse the water consumption for the area. The amount of water coming into the treatment centre, the amount output and amount consumed can be analysed to determine the peak consumption hours versus the low consumption hours. By determining which hours see the most water use, the habits and routines of people can be analyzed. These low use times will also indicate the best times for maintenance and testing to be conducted. The areas which consume the most water can also be determined. The volume of water flowing through certain sections and buildings can be analyzed to determine which locations and sections of the network use the most water. This data can also be compared to previous years to determine if there has been an increase or decrease in water consumption and changes in peak hours for the area and if it relates to population changes. For example, four new houses are built along the network, increasing the population by 12. After this, the water consumption goes up. This can then be used to create a projection for the future which can help to identify where changes in the current network may need to happen to be able to keep up with the growing demand for drinking water.

References

- 1 - National Research Council. 1982. *Drinking Water and Health, Volume 4*. Chapter II, Elements of Public Water Supplies. Washington, DC: The National Academies Press. [Elements of Public Water Supplies - Drinking Water and Health - NCBI Bookshelf \(nih.gov\)](#)
- 2 - Centers for Disease Control and Prevention. (2022, May 16). *Water treatment*. Centers for Disease Control and Prevention. Retrieved April 14, 2023, from https://www.cdc.gov/healthywater/drinking/public/water_treatment.html#:~:text=water%20for%20drinking-,Disinfection,parasites%2C%20bacteria%2C%20or%20viruses.
- 3 - Nathanson, J. A. (2023, April 4). *water supply system*. *Encyclopedia Britannica*. <https://www.britannica.com/technology/water-supply-system>
- 4 - City of Toronto. (2022, November 24). *Watermains*. City of Toronto. Retrieved April 14, 2023, from <https://www.toronto.ca/services-payments/building-construction/infrastructure-city-construction/understanding-city-construction/water-sewer-mains/>
- 4 - St. Johns Department of Works, Water and Wastewater Division. (n.d.). *Water Main Construction Techniques*. Retrieved April 15, 2023, from <https://www.gov.nl.ca/ecc/files/waterres-training-adww-2014-05-jason-phillips.pdf>
- 5 - QED Environmental Systems. (n.d.). *How To Detect Underground An Underground Water Leak*. Retrieved April 14, 2023, from <https://www.qedenv.com/en-us/markets-solutions/water-remediation-and-treatment/water-leak-detection/how-to-detect-an-underground-water-leak/>
- 6 - Krapez, J.-C., Sanchis Muñoz, J., Mazel, C., Chatelard, C., Déliot, P., Frédéric, Y.-M., Barillot, P., et al. (2022). Multispectral Optical Remote Sensing for Water-Leak Detection. *Sensors*, 22(3), 1057. MDPI AG. Retrieved from [Sensors | Free Full-Text | Multispectral Optical Remote Sensing for Water-Leak Detection \(mdpi.com\)](#)